

Iterated Rationality Models and conjunctive readings of disjunctions*

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1. Introduction

Scalar Implicatures (SIs), such as the inference from ‘some’ to ‘some but not all’ in (1), have been extensively studied in recent decades, both theoretically and experimentally.

- (1) John did some of the homework
 $\rightsquigarrow$ John did some but not all of the homework

In the pursuit of a theory that accounts for SIs, two main approaches have been developed: (a) the *pragmatic approach*, according to which SIs are a pragmatic phenomenon that arises at the speech act level, driven by conversational principles (Horn 1972, Grice 1989, a.o.); and (b) the *grammatical approach*, according to which SIs are logical entailments derived compositionally within grammar, usually by a covert exhaustivity operator (Fox 2007, Bar-Lev and Fox 2017, 2020, a.o.). The present work focuses on a specific subgroup of pragmatic theories, referred to as *Iterated Rationality Models* (IRMs; Benz 2006, Benz and Van Rooij 2007, Franke 2009, 2011, Frank and Goodman 2012, Rothschild 2013, Bergen et al. 2016, a.o.). In recent work, Fox and Katzir (2021) compare the two approaches and present two arguments in favor of a grammatical approach to SIs, based on conjunctive readings of disjunctive sentences. First, they argue that IRMs fail to derive *conjunctive readings of disjunctions* (CRDs) with more than two disjuncts. Second, they argue that SI computation is modular, which is at odds with the pragmatic nature of IRMs. The present work proposes to rethink the initial (naive) speaker assumption to deal with CRDs, offering a simple and general solution to this failure of existing IRMs by replacing this assumption with a more nuanced notion of speaker. These results undercut Fox and Katzir’s (2021) first argument against IRMs, based on the inability to provide a general derivation of CRDs.

In section 2, I start by describing the grammatical and IRM approaches and their prediction for CRDs in more detail, and specifying the challenge faced by IRMs. In section 3,

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I present my proposal, explain the intuition behind it, and show how it overcomes this challenge. In section 4, I address other challenges and questions that warrant further research.

2. Background

2.1 Conjunctive Readings of Disjunctions (CRDs)

Disjunction gives rise to a conjunctive interpretation in multiple configurations where the alternatives are not closed under conjunction (i.e., ‘A’ and ‘B’ are alternatives, but ‘A and B’ is not). Such inferences have been argued to account for the behavior of Warlpiri connectives, (Bowler 2014), disjunction in child language (Singh et al. 2016) and Free Choice disjunctions (Kamp 1974) like (2).

- (2) John is allowed to eat an apple or a banana
- a. $\rightsquigarrow$ John is allowed to eat an apple
 - b. $\rightsquigarrow$ John is allowed to eat a banana

This observation generalizes to any $n \geq 2$ disjunctions within such configurations, and to k -out-of- n disjunctions, i.e., disjunctions of $A_1, \dots, A_k$ given that $A_1, \dots, A_n$ are salient in the context. In these cases, the disjunction gives rise to a conjunctive inference of the disjuncts and an exhaustive inference that excludes the other salient alternatives. For instance, if there are 3 options for dessert – apples, bananas, and cherries – the inferences are:

- (3) John is allowed to eat an apple or a banana
- a. $\rightsquigarrow$ John is allowed to eat an apple
 - b. $\rightsquigarrow$ John is allowed to eat a banana
 - c. $\rightsquigarrow$ John is not allowed to eat cherries

Such inferences have been argued to be SIs (Kratzer and Shimoyama 2002, Alonso-Ovalle 2005), and therefore we can formulate the following desideratum for theories of SIs:

- (4) DESIDERATUM FOR THEORIES OF SCALAR IMPLICATURES: for any $1 \leq k \leq n$, a disjunction ‘ A_1 or ... or A_k ’ with $A_{k+1}, \dots, A_n$ as other salient alternatives, derive:
- a. A conjunctive inference of k -out-of- n $\bigwedge \{A_1, \dots, A_k\}$
 - b. An exhaustive inference of $n-k$ -out-of- n $\bigwedge \{\neg A_{k+1}, \dots, \neg A_n\}$

Focusing on the basic (i.e., 2-way disjunctions such as (2)), CRDs have served as a central criterion for comparing the grammatical and pragmatic approaches: while accounted for under the grammatical theories, they were a major problem for early pragmatic theories, and as such were a key argument for the former in the literature (Fox 2007). However, a prominent body of work on Iterated Rationality Models (IRMs; following the terminology of Fox and Katzir 2021) later offered a pragmatic derivation of these inferences (Franke 2009, van Rooij 2010).

2.2 Iterated Rationality Models (IRMs)

The IRM approach bundles a variety of models that incorporate an iterative process of pragmatic and probabilistic reasoning, including the Rational Speech Act model (Frank and Goodman 2012, Bergen et al. 2016), Iterated Best Response (Franke 2009, 2011), Bidirectional OT (Blutner 1998, 2000, Jäger 2002) a.o. Unlike previous pragmatic theories, some IRMs (Franke 2009, van Rooij 2010) derive the conjunctive reading in cases like (2) (see Fox and Katzir 2021). Simplifying somewhat, under such IRMs the speaker starts by assigning probabilities to every message $m \in M$, given the epistemic state they are in $t \in \Pi$, $P(m|t)$, where M is the set of all possible messages and Π is the partition of the context set into states, which is induced by M . These probabilities are computed based on a *naive speaker assumption*:

$$(5) \quad \text{NAIVE SPEAKER: If } n \text{ messages make } t \text{ true, then: } P(m|t) = \begin{cases} 1/n & \text{if } m(t) = 1 \\ 0 & \text{otherwise} \end{cases}$$

A message identifies a state if its conditional probability is the strict maximum compared to all other messages. The identification process is iterative - in each iteration the listener pairs together messages and states according to this criterion. After being identified, those messages and states are eliminated, and the next iteration is applied to the remaining messages and states. Once convergence has been reached, or if no identification and elimination is possible, the process ends. Then, the listener has to compare the likelihoods of the different states given the message they heard, $P(t|m)$. Using Bayesian reasoning under the assumption that all prior probabilities $P(t)$ are flat, the listener instead compares the values of $P(m|t)$ for various states t given the message m they hear.¹

Consider, for instance, the simple SI of ‘some’ strengthened to ‘some but not all’ as in (1). In that case, the possible messages, or *alternatives* are $M = \{\text{‘some’}, \text{‘all’}\}$. Given these alternatives, the relevant epistemic states that the speaker may convey are that John did all of the homework ($\forall$), that he did some but not all of the homework ($\exists \wedge \neg \forall$), and that he didn’t do any of the homework ($\neg \exists$). In the state $\exists \wedge \neg \forall$, the only consistent message is ‘some’, while in the state $\forall$ both ‘some’ and ‘all’ are consistent. Therefore, according to (5) a naive speaker will pronounce each message with the following probabilities:²

(6) Naive speaker’s probability assignment (‘some’/‘all’):

$P(m t)$	‘some’	‘all’
$\exists \wedge \neg \forall$	1	0
$\forall$	$\frac{1}{2}$	$\frac{1}{2}$

¹According to Bayes rule, $P(t|m) = \frac{P(m|t) \cdot P(t)}{P(m)}$. Therefore, $P(t|m) > P(t'|m)$ can be rewritten as $\frac{P(m|t) \cdot P(t)}{P(m)} > \frac{P(m|t') \cdot P(t')}{P(m)}$. Since we are interested in relative comparison of probabilities, the denominator $P(m)$ can be omitted from both sides. Assuming flat priors, $P(t)$ and $P(t')$ can also be ignored.

²For ease of presentation, I omit from the probabilities tables rows in which all probabilities are 0. E.g. in (6), the row of the state $\neg \exists$ is omitted, since its probabilities are all 0 (no message is compatible with that state).

Given these probabilities, the listener infers the meaning of each message within one step: ‘some’ identifies $\exists \wedge \neg \forall$, since $P(\text{‘some’} | \exists \wedge \neg \forall) = 1 > \frac{1}{2} = P(\text{‘some’} | \forall)$; and ‘all’ identifies $\forall$ since $P(\text{‘all’} | \forall) = \frac{1}{2}$ is the only positive probability given the state ‘all’. In particular, ‘some’ is interpreted as ‘some but not all’, as desired.

Notably, the IRM presented here is a simplified model drawn from Fox and Katzir (2021). It incorporates the core assumptions of the Iterated Based Response model of Franke (2009, 2011), and appears to be much farther from other IRMs proposed in the literature (such as the Rational Speech Act framework). However, it is sufficient to use this model to generalize about the IRMs approach, since the present work focuses on CRDs. Most of the literature on IRMs cannot derive CRDs and focuses on simple kinds of SIs (such as (1)). As mentioned by Fox and Katzir (2021), Franke (2009, 2011) and van Rooij (2010) frameworks are the only ones that derive CRDs. Therefore, this simple model offers a relatively clear presentation of the results achieved by IRMs with respect to CRDs.

2.3 IRMs and CRDs: 2 disjuncts

The IRM above derives CRDs in the simple case of 2 disjuncts (like (2)): considering a 2-disjunct sentence ‘A or B’, the alternative messages are ‘A’, ‘B’, ‘A or B’ and the states are $\{\neg A \wedge \neg B, A \wedge \neg B, \neg A \wedge B, A \wedge B\}$. The naive speaker’s probabilities are as in (7). Based on them, in the first step the listener identifies the alternative messages ‘A’ and ‘B’ with $A \wedge \neg B$ and $\neg A \wedge B$ respectively (as indicated in pink in the table). ‘A or B’ cannot identify any state at this stage because there is no single state that obtains the highest probability, but rather there is a tie between $A \wedge \neg B$ and $\neg A \wedge B$. However, these states (and their corresponding messages) are eliminated after the first iteration (as indicated in gray). In the second step, ‘A or B’ is the only remaining message, as is the state $A \wedge B$, so they are necessarily paired together (as indicated in yellow).

(7) Naive speaker’s probability assignment (‘A’/‘B’/‘A or B’):

$P(m t)$	‘A’	‘B’	‘A or B’
$A \wedge \neg B$	$\frac{1}{2}$	0	$\frac{1}{2}$
$\neg A \wedge B$	0	$\frac{1}{2}$	$\frac{1}{2}$
$A \wedge B$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

■ identified message-state pairs in iteration I

■ cells eliminated after iteration I

■ identified message-state pairs in iteration II

2.4 Beyond 2 disjuncts

As formulated in section 2.1, CRDs are not limited to 2-way disjunctions, and generalize to every $n \geq 2$ disjunctions and to k -out-of- n disjunctions, and a theory of SIs has to derive them all as formulated in (4). Indeed, this desideratum is obtained within the grammatical approach, in the same way it handles 2 disjuncts (see Fox and Katzir 2021). However, van Rooij (2010), Franke (2011) and later Fox and Katzir (2021) observe that IRMs fails with constructions involving more than 2 disjuncts, even though they appear to be a simple extension. Consider the case of 3 disjuncts, ‘A or B or C’ that should be interpreted as a

conjunction ($A \wedge B \wedge C$) given the desideratum in (4). The alternatives in that case are $\{‘A’, ‘B’, ‘C’, ‘A \text{ or } B’, ‘A \text{ or } C’, ‘B \text{ or } C’, ‘A \text{ or } B \text{ or } C’\}$, and the probabilities assigned by the naive speaker are as in (8). Based on these probabilities, in the first step the listener identifies the alternative messages ‘A’, ‘B’, ‘C’ with $A \wedge \neg B \wedge \neg C$, $\neg A \wedge B \wedge \neg C$ and $\neg A \wedge \neg B \wedge C$ respectively (as indicated in pink in the table). After eliminating these messages and states (indicated in gray), the remaining states still involve a tie for the first place by multiple messages (see the remaining, white cells in the table). Hence, in the second step, no identification is possible and the procedure ends, and the remaining messages do not get the strengthened conjunctive reading.

(8) Naive speaker’s probability assignment (three disjuncts):

$P(m t)$	‘A’	‘B’	‘C’	‘A or B’	‘A or C’	‘B or C’	‘A or B or C’
$A \wedge \neg B \wedge \neg C$	$\frac{1}{4}$	0	0	$\frac{1}{4}$	$\frac{1}{4}$	0	$\frac{1}{4}$
$\neg A \wedge B \wedge \neg C$	0	$\frac{1}{4}$	0	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{4}$
$\neg A \wedge \neg B \wedge C$	0	0	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$A \wedge B \wedge \neg C$	$\frac{1}{6}$	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$A \wedge \neg B \wedge C$	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$\neg A \wedge B \wedge C$	0	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$A \wedge B \wedge C$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$

■ identified message-state pairs in iteration I ■ cells eliminated after iteration I
□ cells not eliminated after iteration I

Fox and Katzir (2021) observe that this IRM fails twice: neither the full message ‘A or B or C’ nor the partial 2-way disjunctions when there is a third salient alternative (e.g. ‘A or B’ when ‘C’ is salient) identify any state. That is, this IRM fails to satisfy the desideratum in (4). These failures are shared by all existing IRMs and attempts to fix them at most address the former only (van Rooij 2010, Franke 2011). Fox and Katzir (2021) conclude that these results constitute an argument in favor of the grammatical approach over IRMs.

3. Proposal

3.1 IRM with weighted probabilities

The question underlying this work is whether a more nuanced IRM can address the challenge facing existing IRMs. Specifically, it proposes to re-examine the assumption of a naive speaker in (5). Such an assumption – although simple – seems non-trivial in some scenarios. For instance, consider again the of 2-out-of-3 disjuncts case, e.g., the message ‘A or B’ that should be interpreted as the state $A \wedge B \wedge \neg C$. There are multiple messages that are compatible with this state, in particular ‘A or B’ and ‘A or C’. Since both are true in that state, the probability of a naive speaker uttering them is positive and equal – neither of them is considered better than the other. However, focusing on these 2-disjunct messages, one may expect the message ‘A or B’ to be better to convey this state than the message ‘A

or C' – while both disjuncts in the first message are true in the state, the second message has a disjunct which is false in the state (C), and also doesn't have a disjunct which is true in the state (B). In some intuitive sense then, the first message is a better fit for the state than the second, but this is not reflected in the assignment of probabilities according to the naive speaker assumption.

This example suggests a different type of speaker. Instead of a uniform distribution among all messages compatible with a certain state, I propose using weighted probabilities – which intuitively reflect the degree of “overlap” between messages and states – rather than a binary value indicating only whether they are consistent or not. The main question arising is how such “overlap” should be defined and measured. To answer this question, we will follow the intuition behind the 2-out-of-3 case presented above. We have considered ‘A or B’ as a better, or more “overlapping” message than ‘A or C’ with respect to $A \wedge B \wedge \neg C$ because of the consistency of their disjuncts with this state. The more disjuncts are compatible with a state, the more the whole utterance captures that state, and is therefore perceived as corresponding to it. Hence, we would expect that each different disjunct of the message will contribute to the weight, according to its (in)consistency with the given state. Moreover, we would assume that the speaker has no preference among the messages consistent with that state, so that each should contribute equally to the weight.

In order to have some measure which increases with the number of the true disjuncts in a state, we need some mechanism by which we can access the disjuncts composing a given disjunction, and then test which of them are true in that state. For that purpose, I adopt the notion of *deletion alternatives*, Alt_{del} , based on Katzir's (2007) definition of alternatives.³ According to this definition, the deletion alternatives to a disjunction are the disjunction itself and all the partial disjunctions composed by at least one individual disjunct. For instance, $Alt_{del}(\text{'A or B'}) = \{\text{'A'}, \text{'B'}, \text{'A or B'}\}$ and $Alt_{del}(\text{'A'}) = \{\text{'A'}\}$. Indeed, such a set includes also messages other than the individual disjuncts. Hence, the weight of a message m given a state t , $w_t(m)$, will be defined as the number of m 's deletion alternatives that are consistent with t (9a) if m is consistent with t (and 0 otherwise). The probabilities of m given t is then distributed based on these weights (9b).

³It is important to note that we use only deletion alternatives, and not alternatives in general, which include, besides deletion alternatives, alternatives obtained by substitution. Otherwise, by both structural substitution and deletion, all messages with the same number of disjuncts have the same alternatives. For example, the alternatives to ‘A or B’, ‘A or C’ and ‘B or C’ are $\{\text{'A or B'}, \text{'A or C'}, \text{'B or C'}, \text{'A'}, \text{'B'}, \text{'C'}\}$. A weight function based on such alternatives eliminates the effect of weights on distinguishing between these messages, since all are computed based on the compatibility of the same set of alternatives with the same state. In the 2-out-of-3 case, it means that ‘A or B’, ‘A or C’ and ‘B or C’ all have the same weights for $A \wedge B \wedge \neg C$, for example. Accordingly, their probabilities are equal. That is, it brings us back to the initial obstacle, that probabilities do not reflect how close messages are to a state if they are messages of the same complexity (i.e., have the same number of disjuncts). However, though it is necessary to use deletion alternatives in this part, it is not trivial since the model already computes all alternatives to the pronounced utterance, as the set of possible messages that induces the set of states. That is, the IRM computes two different sets of alternatives – one for the measure function and one for identifying states. At this point, I simply make this stipulation, leaving a justification of this choice to future work.

- (9) a. **WEIGHTED MESSAGE SCORES:**
 $w_t(m) = |\{m' : m' \in Alt_{del}(m) \wedge m'(t) = 1\}| \cdot m(t)$
- b. **WEIGHTED PROBABILITIES:** $P(m|t) = w_t(m) / \sum_{m' \in M} w_t(m')$

In other words, the current speaker, unlike the naive speaker, prefers some messages over others to convey a certain state, since they have a greater “overlap” with this state in terms of the number of true deletion alternatives, thus considered better.

3.2 Predictions for CRDs

Before proceeding to the predictions of this revised IRM for CRDs, let us start with the simple SI of ‘some’ strengthened to ‘some but not all’ (like (1)). The alternatives in this case, {‘some’, ‘all’}, are obtained by substitution. That is, the deletion alternatives to each message include the message itself only. It means that for a given message and state, the weight is 1 or 0 and indicates whether the message is compatible with the state or not. With that binary weighting, the probabilities are similar to those computed by the original IRM (i.e., (6)). As shown in section 2.2, the original IRM derives the strengthen inference as desired. Hence, the revised IRM derives it as well.

Now, consider the basic case of CRD with two conjuncts, like (2). The deletion alternatives to ‘A’ and ‘B’ are {‘A’} and {‘B’} respectively. Therefore, the weights assigned for each of them according to (9a) are 0 or 1, indicating whether they are compatible with a certain state or not. For example, $w_{A \wedge \neg B}(\text{‘A’}) = 1$ and $w_{A \wedge \neg B}(\text{‘B’}) = 0$. However, $Alt_{del}(\text{‘A or B’}) = \{\text{‘A’}, \text{‘B’}, \text{‘A or B’}\}$, so its weights are natural numbers on a scale of 0 to 3. For example, $w_{A \wedge \neg B}(\text{‘A or B’}) = 2$ (because the deletion alternatives ‘A’ and ‘A or B’ are the only ones compatible with that state) and $w_{A \wedge B}(\text{‘A or B’}) = 3$. Table (10) summarizes these weights (the last column shows the total weight for each state, according to which the probabilities are computed).

- (10) **Weighted Message Scores (‘A’/‘B’/‘A or B’):**

$w_m(t)$	‘A’	‘B’	‘A or B’	Σ
$A \wedge \neg B$	1	0	2	3
$\neg A \wedge B$	0	1	2	3
$A \wedge B$	1	1	3	5

Based on these weights and according to (9b), the probabilities are as in (11). The revised IRM still derives a CRD: as in the previous model (see (7)), in the first iteration ‘A’ identifies $A \wedge \neg B$ and ‘B’ identifies $\neg A \wedge B$. After elimination, ‘A or B’ identifies $A \wedge B$ (since they are the only remaining message and state), as desired.

(11) Weighted probabilities ('A'/'B'/'A or B'):

$P(m t)$	'A'	'B'	'A or B'
$A \wedge \neg B$	$\frac{1}{3}$	0	$\frac{2}{3}$
$\neg A \wedge B$	0	$\frac{1}{3}$	$\frac{2}{3}$
$A \wedge B$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{3}{5}$

■ identified message-state pairs in iteration I
■ cells eliminated after iteration I
■ identified message-state pairs in iteration II

We move on to the 3-disjunct case, in which the previous model failed to achieve the desideratum both in the 2-out-of-3 and 3-out-of-3 cases as we have seen in (8). As before, the only deletion alternative to an individual disjunct is itself, and its weight is therefore binary; and the deletion alternatives to a 2-way disjunction include itself and its individual disjuncts, so its weights are between 0 to 3. As for the full message, $Alt_{del}(\text{'A or B or C'}) = \{\text{'A'}, \text{'B'}, \text{'C'}, \text{'A or B'}, \text{'A or C'}, \text{'B or C'}, \text{'A or B or C'}\}$ so its weights are natural numbers between 0 to 7. Hence, based on (9), the probabilities are as in (12). In the first step 'A', 'B', and 'C' identify $A \wedge \neg B \wedge \neg C$, $\neg A \wedge B \wedge \neg C$ and $\neg A \wedge \neg B \wedge C$ (resp.; indicated in pink). After eliminating these messages and states (as indicated in light gray), in the second step, 'A or B', 'B or C', 'A or C' identify $A \wedge B \wedge \neg C$, $\neg A \wedge B \wedge C$ and $A \wedge \neg B \wedge C$ (resp.; indicated in yellow). After further elimination (indicated in dark gray), 'A or B or C' is the only remaining message as is the state $A \wedge B \wedge C$, so they are necessarily paired together (indicated in green).

(12) Weighted probabilities (three disjuncts):

$P(m t)$	'A'	'B'	'C'	'A or B'	'A or C'	'B or C'	'A or B or C'
$A \wedge \neg B \wedge \neg C$	$\frac{1}{9}$	0	0	$\frac{2}{9}$	$\frac{2}{9}$	0	$\frac{4}{9}$
$\neg A \wedge B \wedge \neg C$	0	$\frac{1}{9}$	0	$\frac{2}{9}$	0	$\frac{2}{9}$	$\frac{4}{9}$
$\neg A \wedge \neg B \wedge C$	0	0	$\frac{1}{9}$	0	$\frac{2}{9}$	$\frac{2}{9}$	$\frac{4}{9}$
$A \wedge B \wedge \neg C$	$\frac{1}{15}$	$\frac{1}{15}$	0	$\frac{1}{5}$	$\frac{2}{15}$	$\frac{2}{15}$	$\frac{2}{5}$
$A \wedge \neg B \wedge C$	$\frac{1}{15}$	0	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{2}{15}$	$\frac{2}{5}$
$\neg A \wedge B \wedge C$	0	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{2}{5}$
$A \wedge B \wedge C$	$\frac{1}{19}$	$\frac{1}{19}$	$\frac{1}{19}$	$\frac{3}{19}$	$\frac{3}{19}$	$\frac{3}{19}$	$\frac{7}{19}$

■ identified message-state pairs in iteration I ■ cells eliminated after iteration I
■ identified message-state pairs in iteration II ■ cells eliminated after iteration II
■ identified message-state pairs in iteration III

On top of 3-disjunctions, these results generalize to any $n \geq 2$ disjunctions, both for n -out-of- n and k -out-of- n cases,⁴ thus achieving the desideratum in (4) and significantly broadening the scope of the model's success. Hence, the grammatical approach has no empirical advantage over this IRM with respect to CRDs.

⁴A full proof is in the appendix of Frischhoff (to appear).

4. Conclusion

In this work, I presented a new conception of a (non-naive) speaker in the IRM approach, that is based on an intuition of ‘degree of overlap’ between messages and epistemic states, rather than reflecting only whether they are compatible or not. I showed that an IRM incorporating this perspective succeeds in deriving CRDs beyond the basic case of 2 disjuncts – in contrast to the IRMs proposed in the literature, as observed by Franke (2009, 2011), van Rooij (2010) and Fox and Katzir (2021). By doing so, this work eliminates Fox and Katzir’s (2021) argument against IRMs based on CRDs. Alongside this achievement, there are still several issues to address in the evaluation of this proposal. First, as briefly discussed in fn. 4, the use of different sets of alternatives – one for determining states and one for measuring weights – is non-trivial. Therefore, this point should be further explored when considering the current proposal as a comprehensive theory of SIs. Second, in section 3.1 I described the weights as conveying an intuition of “overlapping” messages and states in terms of individual disjuncts. However, one could propose other weight functions that may be no less natural. A more grounded motivation to choose such a weight function can therefore help in its evaluation. Moreover, there are concerns raised about IRMs that the proposed model does not alleviate. One such concern is that this IRM, like other IRMs, is influenced by actual priors. Fox and Katzir (2021) point out that prior sensitivity is another challenge for a theory of SIs (see also Cremers et al. 2022). A second concern is related to an argument made by Asherov et al. (2024) regarding IRMs and inferences in reference games. They show that unlike grammatical approaches, IRMs derive unattested inferences in such games; the revised IRMs proposed here suffers from the same problem. To conclude, the present work renders IRMs a potentially viable theory of SIs, to the extent that other concerns for IRMs, as the one mentioned above, can be eliminated as well.

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